2 **(a)** Show that the system of equations

$$x - y + 2z = 4,$$
$$x - y - 3z = a,$$
$$x - y + 7z = 13,$$

where a is a constant, does not have a unique solution. [2]

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(b) Given that $a = -5$, show that the system of equations in part **(a)** is consistent. Interpret this situation geometrically. [3]

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(c) Given instead that $a \neq -5$, show that the system of equations in part **(a)** is inconsistent. Interpret this situation geometrically. [2]

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